

# **↳ PART 1: Enable Internet on BeagleBone Black via USB**

## **◆ Step A: Assign Correct Static IPs**

*On Ubuntu (Host):*

*Use interface enx415f6f3937f, and assign 192.168.6.1:*

```
sudo ip addr flush dev enx415f6f3937f
sudo ip addr add 192.168.6.1/24 dev enx415f6f3937f
sudo ip link set enx415f6f3937f up
```

*On BBB (via serial or SSH):*

```
sudo ip addr flush dev usb0
sudo ip addr add 192.168.6.2/24 dev usb0
sudo ip link set usb0 up
sudo ip route add default via 192.168.6.1
```

## **◆ Step B: Set DNS on BBB**

```
echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf
```

## **◆ Step C: Enable NAT on Ubuntu (Internet Sharing)**

*1. Enable IP forwarding:*

```
sudo sysctl -w net.ipv4.ip_forward=1
echo "net.ipv4.ip_forward=1" | sudo tee -a
/etc/sysctl.conf
```

2. Setup NAT (Replace wlp0s20f3 with your internet interface):

```
sudo iptables -t nat -A POSTROUTING -o wlp0s20f3 -j MASQUERADE  
sudo iptables -A FORWARD -i enxe415f6f3937f -o wlp0s20f3 -j ACCEPT  
sudo iptables -A FORWARD -i wlp0s20f3 -o enxe415f6f3937f -m state --state RELATED,ESTABLISHED -j ACCEPT
```

## ◆ Step D: Test Internet on BBB

On BBB:

```
ping 8.8.8.8  
ping google.com
```

If both succeed, internet is working! 2

## **x PART 2: Enable & Test CAN Interface on BBB**

### ◆ Step A: Enable CAN Pins in Device Tree

On BBB, edit /boot/uEnv.txt:

```
sudo nano /boot/uEnv.txt
```

Uncomment or add:

```
enable_uboot_overlays=1
uboot_overlay_addr4=/lib/firmware/BB-CAN0-00A0.dtbo
uboot_overlay_addr5=/lib/firmware/BB-CAN1-00A0.dtbo
```

Then reboot:

```
sudo reboot
```

## ◆ Step B: Bring Up CAN Interfaces

After reboot:

```
sudo ip link set can0 up type can bitrate 500000
sudo ip link set can0 up
```

To verify:

```
ifconfig can0
```

To bring CAN1 up (if used):

```
sudo ip link set can1 up type can bitrate 500000
sudo ip link set can1 up
```

## ◆ Step C: Send & Receive CAN Messages (Loopback test)

Enable loopback for test:

```
sudo ip link set can0 down
sudo ip link set can0 up type can bitrate 500000 loopback
on
```

Open two terminals and run:

Terminal 1 (receive):

candump can0

Terminal 2 (send):

cansend can0 123#1122334455667788

You should see the message appear in Terminal 1